A PROJECT TITLE

ON

Secure CI/CD for AI Services using GitHub Actions,Trivy and tfsec

GENERATIVE AI

BACHELOR OF TECHNOLOGY

in

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by

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Department Of Computer Science and Engineering

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1. Abstract:

The project titled "Secure CI/CD for AI Services using GitHub Actions, Trivy and tfsec" focuses on developing a secure and automated process for building and managing Generative AI applications. A Generative AI-based Student Chatbot is considered as an example application to demonstrate the proposed system. The chatbot is developed using Python and FastAPI and integrated with the Gemini API to generate intelligent responses to user queries.

Docker is used to package the chatbot along with its required libraries and dependencies, ensuring that the application runs consistently across different environments. GitHub is used to store and manage the project source code, while GitHub Actions automates the Continuous Integration and Continuous Deployment (CI/CD) process. It automatically executes the configured workflow whenever changes are pushed to the repository.

To improve application security, Trivy is integrated to scan application dependencies and Docker images for known security vulnerabilities. Additionally, tfsec is used to examine Terraform configuration files and identify potential security misconfigurations. These security tools help detect possible risks before the application is deployed.

The main objective of this project is to reduce manual effort, identify security issues at an early stage and improve the reliability of AI application development. By combining Generative AI, Docker, GitHub Actions, Trivy and tfsec, the project demonstrates how security checking can be automated and integrated into the development lifecycle of AI-powered applications.

2. INTRODUCTION:

Generative Artificial Intelligence (Generative AI) is a technology that can generate human-like text and provide intelligent responses based on user inputs. AI chatbots are common applications of this technology, helping users obtain information and answers

quickly. However, developing an AI application also requires proper security checking to protect it from potential vulnerabilities.

This project focuses on developing a secure and automated CI/CD pipeline for Generative AI applications using GitHub Actions, Trivy and tfsec. A Student Chatbot integrated with the Gemini API is used as an example application. Python and FastAPI are used to develop the chatbot, allowing users to ask questions and receive AI-generated responses.

Docker is used to package the application and its dependencies into a Docker image. GitHub manages the source code, while GitHub Actions automates the development and security-checking process. Trivy scans application dependencies and Docker images to identify known vulnerabilities, whereas tfsec examines Terraform files to detect insecure infrastructure configurations.

The main aim of this project is to integrate security checking into the AI application development process. It helps identify potential security problems before deployment, reduces manual work and improves the overall reliability of Generative AI applications.

3. SOFTWARE REQUIREMENTS:

1. Operating System: Windows 10/11 or Ubuntu.
2. Programming Language: Python.
3. Backend Framework: FastAPI.
4. AI Service: Google Gemini API.
5. Containerization: Docker.
6. Source Code Management: Git and GitHub.
7. CI/CD Tool: GitHub Actions.
8. Security Tools: Trivy and tfsec.
9. Infrastructure Tool: Terraform
10. Code Editor: Visual Studio Code (VS Code).

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